

DATA MINING TASK – 8

DEMONSTRATE CLASSIFICATION PROCESS ON CUSTOMER.ARFF DATASET USING NAIVE BAYES ALGORITHM.

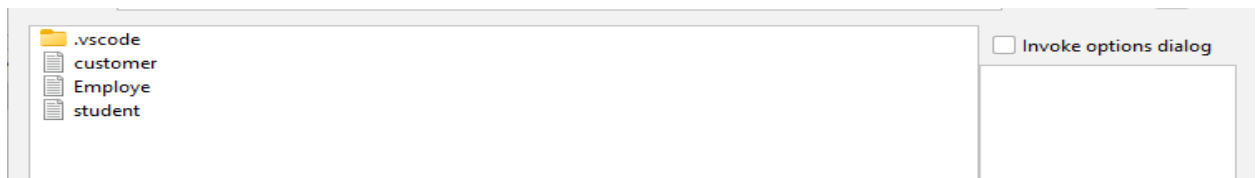
Description: What is Naive Bayes Algorithm

Naive Bayes is a supervised machine learning algorithm it is not a single algorithm it is a family of algorithms. This algorithm is mainly used in classification purpose.

Real-time applications:

The real time applications of naïve bayes are text classification, rating classification and spam filtering etc..

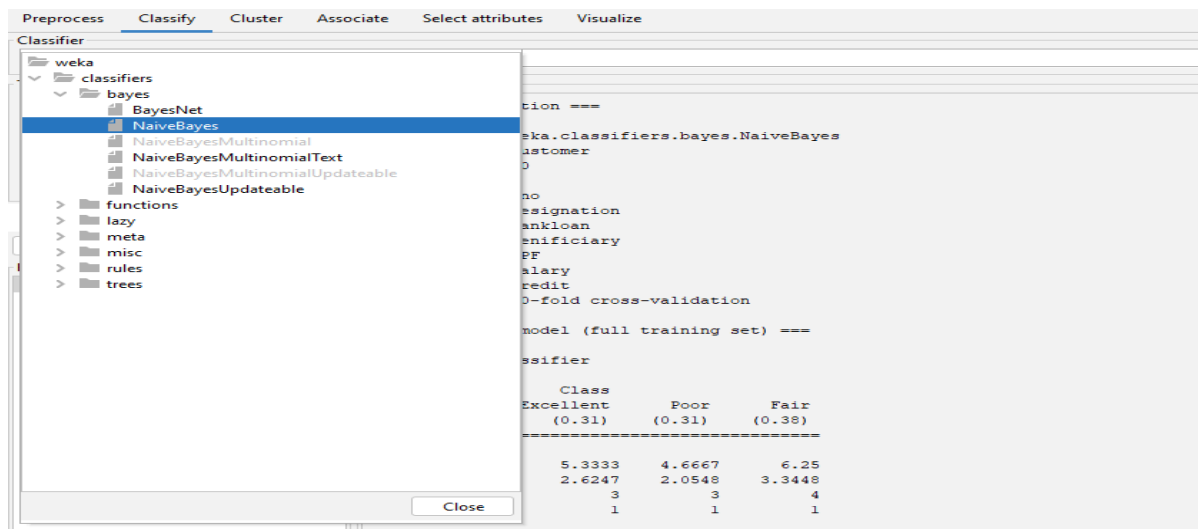
- Here we have taken the previously created dataset which mean customer.arff



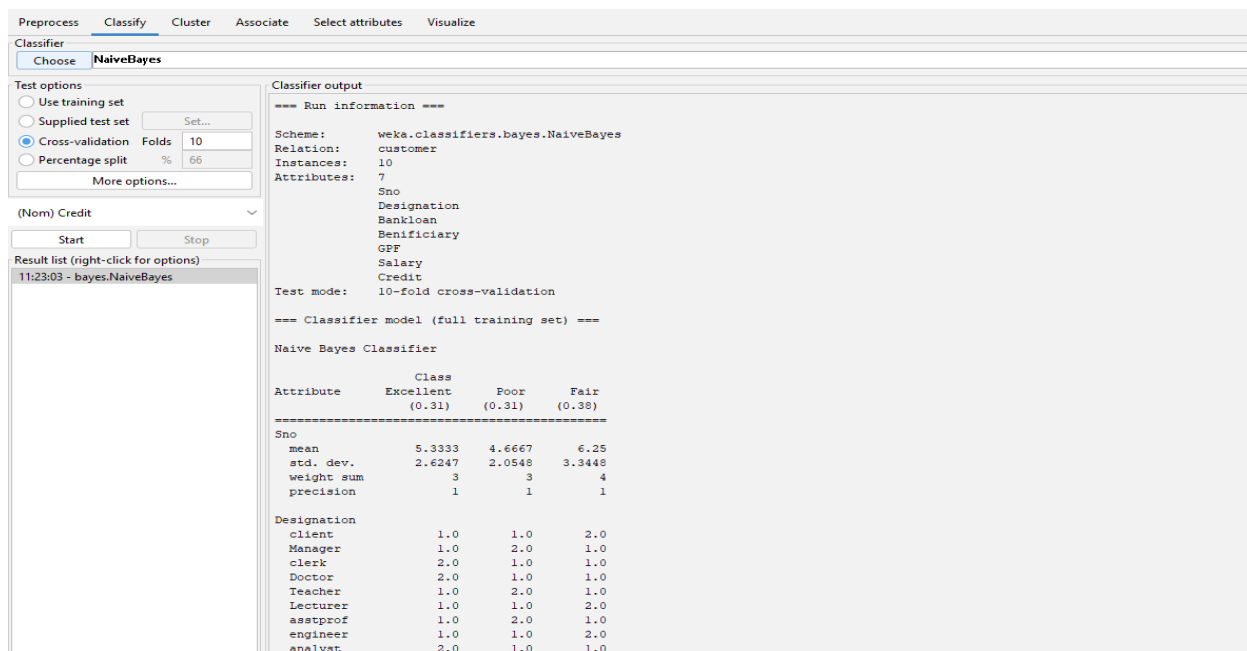
- The data is as follows

Viewer							
Relation: customer							
No.	1: Sno Numeric	2: Designation Nominal	3: Bankloan Nominal	4: Benificiary Nominal	5: GPF Nominal	6: Salary Nominal	7: Credit Nominal
1	1.0	client	Yes	No	Yes	Moder...	Fair
2	2.0	Manager	No	Yes	No	Low	Poor
3	3.0	clerk	Yes	Yes	Yes	High	Excellent
4	4.0	Doctor	No	Yes	Yes	High	Excellent
5	5.0	Teacher	No	No	No	Low	Poor
6	6.0	Lecturer	Yes	No	Yes	Moder...	Fair
7	7.0	asstprof	No	No	Yes	Low	Poor
8	8.0	engineer	Yes	No	No	Moder...	Fair
9	9.0	analyst	No	Yes	Yes	Moder...	Excellent
10	10.0	Proffesor	Yes	No	Yes	High	Fair

- Go to classify and then go to the bayes folder and then on the Naive Bayes



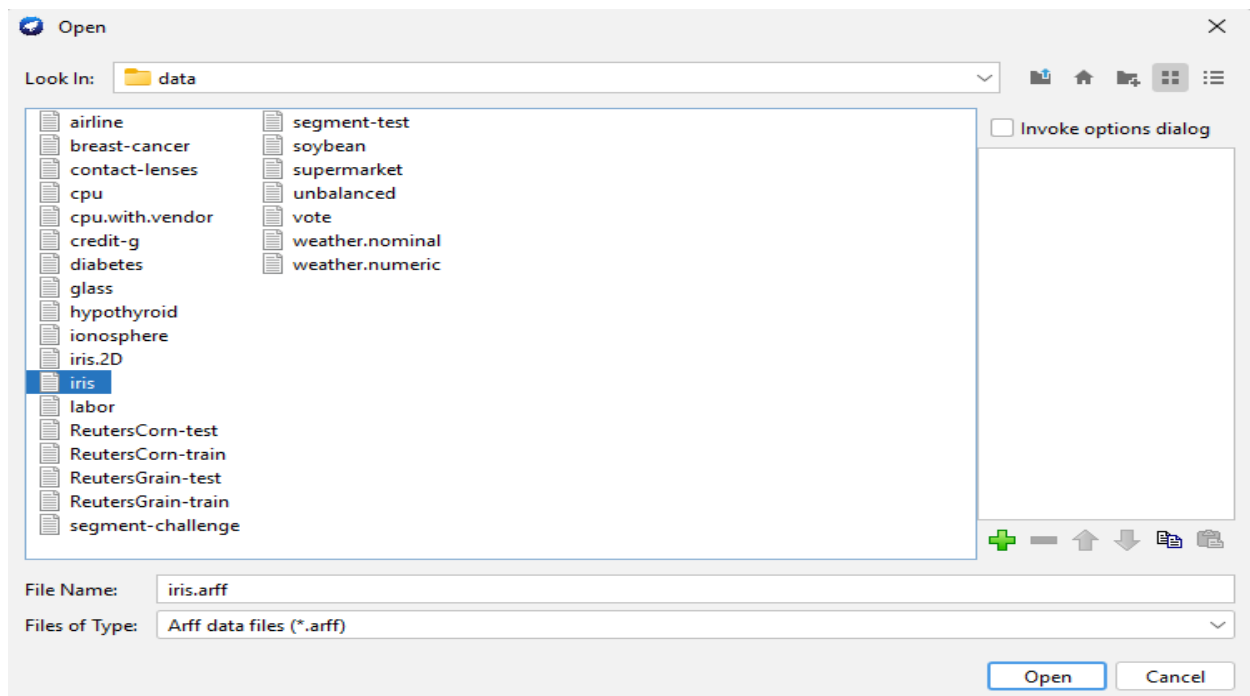
- Then the data is as shown below.



DATA MINING TASK – 9

DEMONSTRATE CLUSTERING PROCESS ON PREDEFINED WEKA DATASET IRIS.ARFF DATASET USING SIMPLEKMEANS ALGORITHM.

- Load the iris.arff dataset.



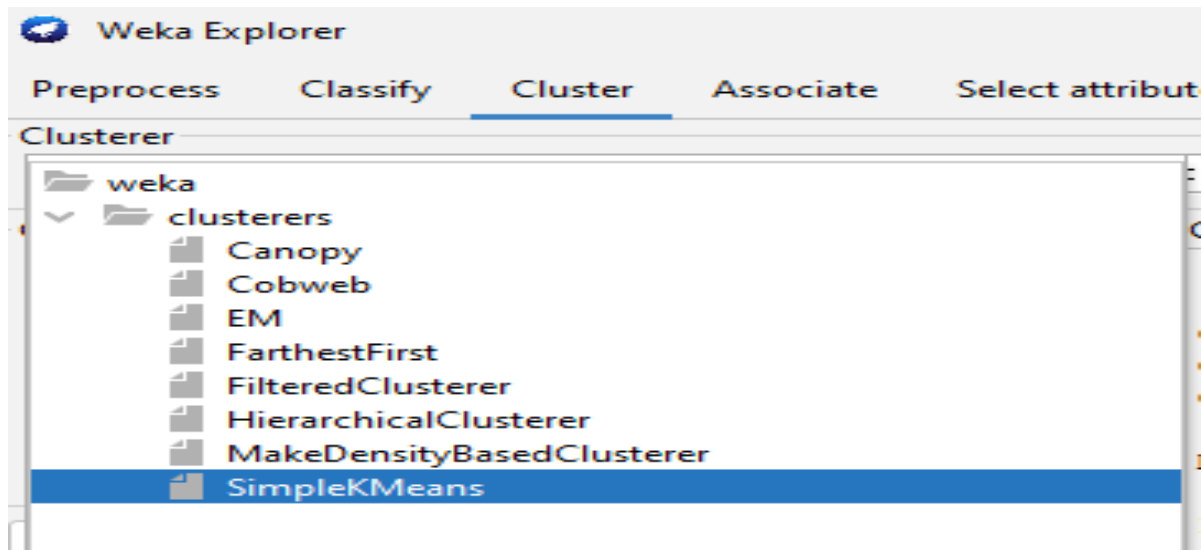
- The dataset is as follows:

Viewer

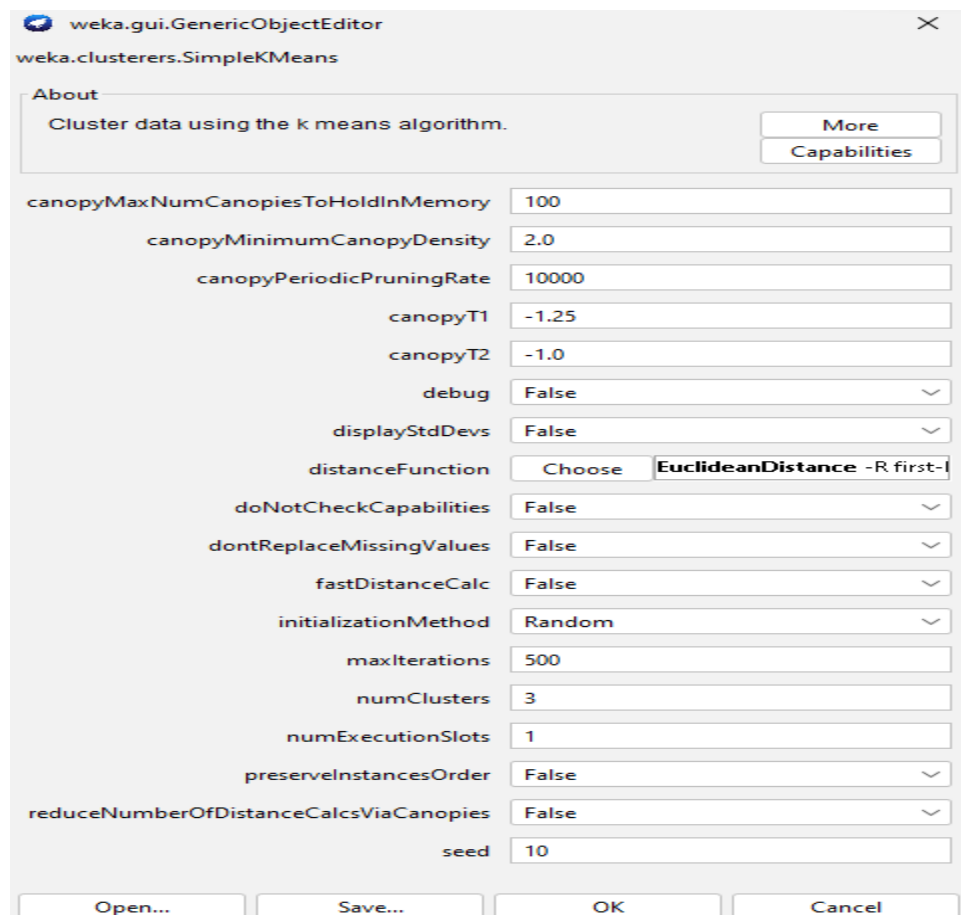
Relation: iris

No.	1: sepallength Numeric	2: sepalwidth Numeric	3: petallength Numeric	4: petalwidth Numeric	5: class Nominal
1	5.1	3.5	1.4	0.2	Iris-se...
2	4.9	3.0	1.4	0.2	Iris-se...
3	4.7	3.2	1.3	0.2	Iris-se...
4	4.6	3.1	1.5	0.2	Iris-se...
5	5.0	3.6	1.4	0.2	Iris-se...
6	5.4	3.9	1.7	0.4	Iris-se...
7	4.6	3.4	1.4	0.3	Iris-se...
8	5.0	3.4	1.5	0.2	Iris-se...
9	4.4	2.9	1.4	0.2	Iris-se...
10	4.9	3.1	1.5	0.1	Iris-se...

- Now click on cluster and then choose SimpleKmeans:



- Then right click on the show properties, there it displays a dialogue box is as follows:



- And then click on ok it display output is as follows

The screenshot shows the Weka GUI with the 'Cluster' tab selected. The 'SimpleKMeans' algorithm is chosen. The 'Cluster mode' section has 'Use training set' selected and 'Store clusters for visualization' checked. The 'Cluster output' pane displays the following information:

kMeans
=====

Number of iterations: 3
Within cluster sum of squared errors: 7.817456892309574

Initial starting points (random):
Cluster 0: 6.1,2.9,4.7,1.4,Iris-versicolor
Cluster 1: 6.2,2.9,4.3,1.3,Iris-versicolor
Cluster 2: 6.9,3.1,5.1,2.3,Iris-virginica

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (150.0)	Cluster# 0 (50.0)	1 (50.0)	2 (50.0)
sepal.length	5.8433	5.936	5.006	6.588
sepal.width	3.054	2.77	3.418	2.974
petal.length	3.7587	4.26	1.464	5.552
petal.width	1.1987	1.326	0.244	2.026
class	Iris-setosa Iris-versicolor	Iris-setosa	Iris-virginica	

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	50 (33%)
1	50 (33%)
2	50 (33%)

- Then right click for option and select Visualize cluster assignments.

The screenshot shows the 'Result list (right-click for options)' pane. A right-click context menu is open, displaying the following options:

- View in main window
- View in separate window
- Save result buffer
- Delete result buffer(s)
- Load model
- Save model
- Re-evaluate model on current test set
- Re-apply this model's configuration
- Visualize cluster assignments
- Visualize tree

- The following graph has been displayed.

